

AI + Photo Storage Cost Analysis

Date: 2026-07-16 · **Scope:** Per-game cost of existing OpenAI features (optimiser, embeddings), the planned "brain dump" field-splitting feature, and the planned R2 photo storage system, plus platform running costs (hosting, analytics, email, domain).

Cost trajectory at a glance

Which lines increase in price as the app grows, and at what point.

Cost	Increases with scale?	Trigger threshold	Notes
Railway (hosting)	Yes	Scales continuously with CPU/RAM load, not a hard step	~\$8/mo early → ~\$20/mo at 10k MAU → ~\$35–80/mo past 25k MAU
Amplitude (analytics)	Yes	~12,500 MAU (~2M events/mo, @160 events/user assumption)	\$0 → flat \$49/mo (Plus) up to ~150k MAU, then custom Growth pricing
Resend (email)	Yes	~3,000 password-reset emails/month	\$0 → \$20/mo (50k emails)
OpenAI (optimiser + embeddings)	No, not materially	Sub-cent per game regardless of volume; 10k games/mo ≈ \$3.50–\$6 total	Rounding error against the above
Cloudflare R2 (photo storage)	No, not materially	Free tier: 10GB storage / 1M writes / 10M reads per month	Would need >10GB of photos or many millions of reads/month to cross — not expected pre-launch
Domain	No	Fixed ~\$1/mo	—

Bottom line for the increases: Railway and Amplitude are the two lines that go up as usage grows — Railway rises gradually with traffic, Amplitude jumps at a fixed MAU threshold. R2 and OpenAI would only increase past scale this app isn't projected to reach for a long time.

Context for the finance report

What the app is: *What's That Game* is a platform for storing and discovering games — users create game entries (name, rules, setup, photos), browse and search a public catalogue, upvote, favourite, and optimise their submissions with AI. Backend is a FastAPI REST API; there is a companion frontend (web + mobile). Two catalogue types exist: user-contributed games and verified games.

Stage — pre-launch. The app **has not launched yet**. There is no live user base and no production traffic history. Therefore:

- **All user counts, event volumes, and game-creation rates in this document are assumptions/projections, not measured data.** They exist to model cost *at hypothetical scale*, not

to report actual spend.

- Current real spend is effectively **only the fixed platform baseline** (Railway hosting + domain), with everything else at \$0 free tier. See "Combined monthly run-rate → Early stage" for the closest thing to a today figure.
- Costs should be read as **step functions**, not linear: most services (Amplitude, Resend, R2, OpenAI) sit at \$0 on free tiers and only begin charging once specific thresholds are crossed (e.g. Amplitude at ~2M events/month). The report should focus on **where those thresholds are** and **what MAU/traffic triggers each step**.

How to use this doc (guidance for the finance agent):

- Separate **fixed platform costs** (scale with traffic/load) from **per-unit costs** (scale with games created). Do not multiply fixed costs by game count.
- Treat the events-per-user (~160/month) and Railway compute figures as the two softest assumptions — flag any projection that hinges on them.
- For a launch budget, the **Early-stage row (~\$9/month)** is the realistic starting burn; the higher rows are contingency planning for adoption that has not happened yet.
- No revenue model is included here — this is a **cost/burn-rate** analysis only.

Pricing used (as of July 2026)

Service	Rate	Source
GPT-4.1-nano input	\$0.10 / 1M tokens	OpenAI Pricing
GPT-4.1-nano output	\$0.40 / 1M tokens	OpenAI Pricing
text-embedding-3-small	\$0.02 / 1M tokens (input only)	Embedding pricing 2026
R2 storage	\$0.015 / GB-month	Cloudflare R2 Pricing
R2 Class A ops (writes/uploads)	\$4.50 / 1M ops	Cloudflare R2 Pricing
R2 Class B ops (reads/retrievals)	\$0.36 / 1M ops	Cloudflare R2 Pricing
R2 egress	\$0 (no egress fees)	Cloudflare R2 Pricing
R2 free tier	10GB storage, 1M Class A ops, 10M Class B ops / month	Cloudflare R2 Pricing

Note on transcription: brain-dump speech input uses the phone's native speech-to-text (client-side, free). No server-side transcription (Whisper / GPT-4o-transcribe) cost applies — the backend only ever receives text.

Assumptions

- Optimiser system prompts averaged from the four templates in `src/utls/prompts.py` (description/objective/setup/rules): ~220 tokens/call average.
- Average user-submitted field text: ~200 tokens in, ~150 tokens out per optimiser call.
- 3 optimiser calls per game (per user's stated usage pattern).
- Embedding: combined field text (~800 tokens), 2 embed calls per game lifecycle (1 create + 1 edit, average).
- Photos: 5 photos uploaded per game (of a 10-photo cap), ~800KB average per photo after client-side compression (5MB is the hard cap, not the typical size).
- Photo views: 100 views/month/game assumed for retrieval cost, modeled both with and without Cloudflare CDN caching in front of the public R2 bucket.
- Brain-dump call: one nano call, ~860 tokens in (combined dump text + extraction system prompt), ~400 tokens out (three split fields).

Per-game cost breakdown

Item	Calc	Cost
Optimiser (3 GPT-4.1-nano calls)	$3 \times (430 \text{ tok in} \times \$0.10/1\text{M} + 150 \text{ tok out} \times \$0.40/1\text{M})$	~\$0.0003
Embedding (2 calls, create + edit)	$2 \times (800 \text{ tok} \times \$0.02/1\text{M})$	~\$0.00003
Brain-dump split (1 GPT-4.1-nano call)	$860 \text{ tok in} \times \$0.10/1\text{M} + 400 \text{ tok out} \times \$0.40/1\text{M}$	~\$0.00025
Photo upload (5× Class A)	$5 \times \$4.50/1\text{M}$	~\$0.00002 (one-time)
Photo storage (5 × 800KB = 4MB)	$0.0039\text{GB} \times \$0.015/\text{GB-mo}$	~\$0.00006 / month
Photo retrieval, no CDN cache (worst case)	$500 \text{ ops/mo} \times \$0.36/1\text{M}$	~\$0.00018 / month
Photo retrieval, with CDN cache (~95% hit)	$25 \text{ ops/mo} \times \$0.36/1\text{M}$	~\$0.00001 / month

Totals per game

Scenario	One-time	Recurring (monthly)
Optimiser + embedding + photos (no brain dump)	~\$0.00035	~\$0.00007–\$0.00024
+ text-based brain dump	~\$0.00060	same as above

Brain dump (text-only, client-side speech recognition) adds roughly the cost of one extra optimiser call — not materially different from the baseline. There is no Whisper/audio cost since transcription happens on-device.

At scale

10,000 games created in a month:

Scenario	Total one-time AI + upload cost
Optimiser + embedding + photo uploads only	~\$3.50
+ text-based brain dump	~\$6.00

Recurring storage/retrieval cost stays inside R2's free tier (10GB storage / 1M Class A / 10M Class B per month) for a very long time at this scale — realistically \$0/month until well past 10,000+ accumulated games with photos.

Platform running costs

The costs above are **per-game unit economics** — they scale with games created. The costs below are **fixed platform / traffic-driven** costs that exist regardless of per-game AI usage. Both are needed for a full run-rate.

Additional pricing used (as of July 2026)

Service	Rate	Source
Railway Hobby plan	\$5 / month (includes \$5 usage credit)	Railway Pricing
Railway Pro plan	\$20 / month per seat (includes \$20 usage credit)	Railway Pricing
Railway CPU	\$20 / vCPU-month (\$0.000463/vCPU/min)	Railway Pricing
Railway RAM	\$10 / GB-month (\$0.000231/GB/min)	Railway Pricing
Railway volume storage	\$0.15 / GB-month	Railway Pricing
Railway network egress	\$0.05 / GB	Railway Pricing
Amplitude Free (Starter)	\$0 — 2M events / month	Amplitude Pricing
Amplitude Plus	\$49 / month — up to 300k MTU or 25M events	Amplitude Pricing
Amplitude Growth	Custom (~\$500–2,000+/mo at scale)	Amplitude Pricing
Resend (email) Free	\$0 — 3,000 emails / month, 100 / day	Resend Pricing
Resend Pro	\$20 / month — 50,000 emails	Resend Pricing
.co.uk domain	~\$10–13 / year (~\$1 / month)	registrar

1. Railway (backend hosting) — fixed monthly cost

Estimate, not measured. Railway CLI login was expired at time of writing, so the compute figure below is modeled from the resource-usage rate card, not pulled from actual billing. For the final finance report, run `railway login` and the real current spend can be read from Railway metrics — actual usage beats a model.

The backend is a single FastAPI service plus a Railway-managed Postgres instance. Both are billed on the same usage model (CPU + RAM + volume + egress), and both draw down the plan's included usage credit before any overage is charged.

Modeled steady-state for a low-traffic app:

Component	Assumed usage	Monthly
FastAPI service (CPU)	~0.1 vCPU avg	~\$2.00
FastAPI service (RAM)	~0.25 GB avg	~\$2.50
Postgres (CPU)	~0.05 vCPU avg	~\$1.00
Postgres (RAM)	~0.25 GB avg	~\$2.50
Postgres volume	~1 GB	~\$0.15
Egress	~2 GB (photos served from R2, not Railway)	~\$0.10
Modeled resource usage		~\$8.25

Effective cost by plan (usage credit applied):

Plan	Subscription	Included credit	Effective monthly
Hobby	\$5	\$5	~\$8.25 usage – \$5 credit + \$5 sub = ~\$8.25
Pro (1 seat)	\$20	\$20	usage (\$8.25) inside \$20 credit = ~\$20

Takeaway: at this traffic level Railway is effectively **~\$8–\$20 / month** and does not scale with game count. It scales with concurrent traffic (CPU/RAM), so it steps up only as active-user load grows. Serving photos from R2 (zero egress) rather than Railway keeps egress negligible here.

2. Amplitude (frontend analytics) — traffic-driven, over time

The org is currently on the **free Starter plan** (verified via API: plan `starter_v3`), which allows **2M events / month** at \$0. Amplitude cost is driven by **event volume** (a function of active users × actions each), **not** by games created — so it grows with adoption, not catalogue size.

Assumption: an active user generates **~40 events/session × ~4 sessions/month ≈ 160 events/user/month** (page views, game views, searches, upvotes, uploads, optimiser use).

Monthly active users (MAU)	Est. monthly events (@160/user)	Amplitude plan	Monthly cost
1,000	160k	Free	\$0
5,000	800k	Free	\$0
~12,500	~2.0M	Free tier ceiling	\$0 (at the limit)
20,000	3.2M	Plus	\$49
100,000	16M	Plus (under 25M)	\$49
150,000+	24M+	Plus → Growth	\$49 → custom

Free → paid crossover: ~12,500 MAU. Below that, analytics are free. The first paid step is a flat **\$49/month** (Plus), which then covers all the way to ~25M events (~150k MAU) before a custom Growth contract is needed.

Sensitivity: the crossover depends entirely on the ~160 events/user/month assumption. If instrumentation is heavier (e.g. ~320 events/user), the free-tier ceiling halves to **~6,250 MAU**. Tune the events-per-user figure once real Amplitude event data exists.

3. Other running costs

Cost	Provider	Rate	Notes
Transactional email	Resend	\$0 free (3k/mo) → \$20/mo (50k)	Password-reset emails (<code>src/services/email.py</code>). Free tier fits until ~3,000 resets/month.
Domain	<code>whatsthatgame.co.uk</code> registrar	~\$1 / month (~\$10–13/yr)	Fixed.
TLS / SSL	Railway	\$0	Included.
Google OAuth	Google	\$0	No cost at this scale.
Cloudflare R2	Cloudflare	~\$0/mo (free tier)	See per-game section above.
OpenAI (optimiser + embeddings)	OpenAI	sub-cent/game	See per-game section above.

Combined monthly run-rate

Bringing fixed + traffic-driven + per-game×volume together. Per-game AI/upload cost is ~\$0.00035–\$0.0006 (see above), so even 10,000 new games/month adds only ~\$3.50–\$6.00 — negligible against fixed costs.

Stage	MAU	Fixed (Railway + domain)	Amplitude	Email	Per-game AI (@ games/mo)	Total / month
Early	~1,000	~\$9	\$0	\$0	~\$0.35 (1k games)	~\$9
Growing	~10,000	~\$20	\$0	\$0	~\$3.50 (10k games)	~\$24
Scaling	~25,000	~\$35 (heavier compute)	\$49	\$20	~\$6 (15k games)	~\$110
Large	~100,000	~\$80 (multi-instance)	\$49	\$20	~\$6	~\$155

Railway compute at the higher stages is a rough scale-up and is the least certain line — confirm with real metrics once login is restored.

Bottom line

- **Per-game AI + storage** (OpenAI, R2) is sub-cent and stays inside free tiers for a long time — not the cost driver.
- **Railway** is the dominant *early* cost: a near-fixed **~\$8–\$20/month** regardless of game count, scaling only with traffic. (Currently modeled — verify with `railway login`.)
- **Amplitude** is **\$0 until ~12,500 MAU**, then a flat **\$49/month** to ~150k MAU. It is the main *analytics* line but only kicks in at real scale.
- **Email (Resend)** and the **domain** are trivial (~\$0–\$21/month combined).
- Realistic total run-rate: **~\$9/month early**, **~\$24/month at 10k MAU**, stepping to **~\$110–\$155/month** only once analytics and heavier compute engage past ~25k MAU.

The brain-dump feature, using client-side speech recognition rather than server-side transcription, does not change this materially — it's roughly one extra cheap LLM call per game, not a new cost category.